

Machine Learning Project Work – Group 26

Aniello Di Meglio

Gennaro Foschillo

Francesco Vittorio Allocca

1 Preliminary Activities

As a preliminary step, our team explored the dataset to understand its structure and identify the relevant information required for the implementation of the machine learning model. This process revealed two relevant characteristics of the dataset: first, the presence of duplicate samples; second, a significant class imbalance.

Although duplicate samples were found in the dataset, we considered this issue to be less significant than class imbalance, since duplicates represented only a very small portion of the data. However, we still conducted specific experiments to assess their potential impact on the model's performance. In contrast, we considered class imbalance to be a much more serious issue, as the underrepresented classes could be penalized during the model training phase. This observation led us to consider the use of weighted cross entropy in the future experiments.

2 Experimental Approach

The training phase was organized as a sequence of controlled experiments. The idea was to start from a simple and reproducible baseline and then modify one main component at a time, such as data augmentation, class weighting, preprocessing strategy or model architecture. This made the comparison between experiments more interpretable, since each performance change could be linked to a specific modification of the pipeline. Transfer learning was adopted because the size of the dataset does not justify training a deep convolutional neural network from scratch.

We decided to use ImageNet pre-trained weights because they provide the model with a strong initial knowledge of general visual patterns, such as shapes, edges, and textures. This allows the training process to start from a more effective point rather than learning everything from scratch. The model was then fine-tuned on our dataset so that this general knowledge could be adapted to the specific task of classifying the eight waste categories.

3 Architecture Selection

At this point, we focused on the selection of an appropriate neural network architecture. Among the available alternatives, we identified three potential candidates: ResNet18, EfficientNet, and MobileNet.

ResNet18 was selected as the initial baseline because it is a standard, stable and computationally moderate convolutional neural network. Its residual structure makes optimization easier and its limited size makes it suitable for validating the training pipeline under the memory constraints of the project.

EfficientNet-B0 was then introduced as a stronger architectural comparison. It follows a different design principle based on compound scaling and is commonly used as an accuracy-efficient model. Therefore, it was a suitable candidate to test whether a more modern architecture could improve the ResNet18 baseline while still satisfying the computational constraints.

MobileNet was initially selected as a lightweight option. However, since we observed that the other two architectures consistently remained well below the required RAM limitations, we decided not to further consider this lightweight alternative.

4 Train/validation split

In the initial phase, we started by conducting experiments using fixed data splits, without applying cross-validation. This choice was mainly motivated by the team’s limited experience in immediately identifying the most effective experimental configurations. This approach allowed us to perform a larger number of experiments in less time, while reserving cross-validation, which is more computationally expensive, for the most significant experiments.

We compared the model’s performance with and without duplicate samples and observed only a negligible difference. This result led us to disregard the duplicate-sample issue during the cross-validation phase. However, for non-cross-validation experiment, we report the split information related to the dataset version in which duplicates were removed.

The dataset was divided into training and validation sets using an approximately 80/20 split that contains 12,394 training images and 3,099 validation images. Furthermore, the split was stratified by class label because the dataset is strongly imbalanced.

Table 1: Train/validation class distribution.

Class	Train	Val
Battery	756 (6.10%)	189 (6.10%)
Clothing	5841 (47.13%)	1460 (47.11%)
Glass	1594 (12.86%)	399 (12.88%)
Metal	614 (4.95%)	154 (4.97%)
Organic	788 (6.36%)	197 (6.36%)
Papery	1553 (12.53%)	388 (12.52%)
Plastic	690 (5.57%)	173 (5.58%)
Undifferentiated	558 (4.50%)	139 (4.49%)
Total	12394	3099

5 First experiment: ResNet18 baseline

The first experiment was designed as a controlled baseline for the whole training pipeline. Before introducing additional components such as data augmentation, class-weighted losses or alternative architectures, a simple and reproducible reference model was needed to verify that preprocessing, training, validation, checkpointing and metric computation were working correctly.

The model was an ImageNet-pretrained ResNet18 adapted to the waste classification task by replacing the original classification layer with a new fully connected layer with 8 output neurons, one for each target class. All layers were fine-tuned, rather than training only the final classifier, because the visual domain of the dataset differs from the original ImageNet classification task in terms of object type, background, resolution and acquisition conditions.

The number of layers to unfreeze during fine-tuning can be considered an important hyperparameter, since it generally depends on the size of the dataset and on how different the target domain is from the original pretraining domain. In our case, we adopted an incremental approach to the number of trainable layers. Starting from a configuration in which only the final classification layer was trained, we progressively increased the number of unfrozen layers following the sequence 0, 4, 6, and finally all layers. This allowed us to compare different levels of adaptation of the pretrained model to the waste classification task.

Table 2: Validation balanced accuracy report

# Unfrozen layers	Best Val. Bal. Acc.
0	0.9214
4	0.9539
6	0.9611

This improvement trend led us to the final decision of unfreezing all layers during fine-tuning.

All images were converted to RGB, and resized to 224×224 , in order to be transformed into tensors, and normalized using the ImageNet mean and standard deviation associated with the pretrained weights. This preprocessing keeps the input format consistent with the pretrained model. The same deterministic preprocessing was applied to both training and validation images. This made the experiment a neutral baseline: later improvements or degradations could be attributed to the specific changes introduced in the following experiments.

The loss function was standard CrossEntropyLoss, which is the natural baseline objective for single-label multi-class classification. No class weights and no sampling strategy were used in this first run. This was intentional: the goal was first to observe how a standard loss behaves on the imbalanced dataset before introducing imbalance-aware corrections.

The model was optimized using AdamW with learning rate 1×10^{-4} and weight decay 1×10^{-4} . AdamW was used as a practical optimizer for fine-tuning, while the learning rate was kept small because all pretrained layers were updated. A larger learning rate could modify the pretrained representation too aggressively. Weight decay was included as a mild regularization term to discourage unnecessarily large weights during full fine-tuning.

A ReduceLROnPlateau scheduler monitored the validation loss and reduced the learning rate when the validation loss stopped improving. This was used as an optimization stabilizer: after the first rapid improvement, smaller learning-rate steps can help refine the solution without continuing with the same update magnitude. The scheduler was driven by validation loss, while checkpoint selection was driven by validation Balanced Accuracy.

Training was allowed for a maximum of 30 epochs, with early stopping patience equal to 10. At the end of each epoch, the model was evaluated on the validation set. The best checkpoint was selected according to validation Balanced Accuracy and stored separately from the last checkpoint. This is important because the best validation performance does not necessarily occur at the final epoch.

The best model was obtained at epoch 12, with validation Balanced Accuracy of 0.9758 and validation accuracy of 0.9864. Training continued until epoch 22 due to the early stopping patience, but no later epoch improved the best Balanced Accuracy. This indicates that the model had already reached its best validation trade-off around epoch 12. In later epochs, the training metrics were almost perfect, while validation performance remained slightly lower and did not improve further. Therefore, longer training alone was not considered a promising direction for improving the model.

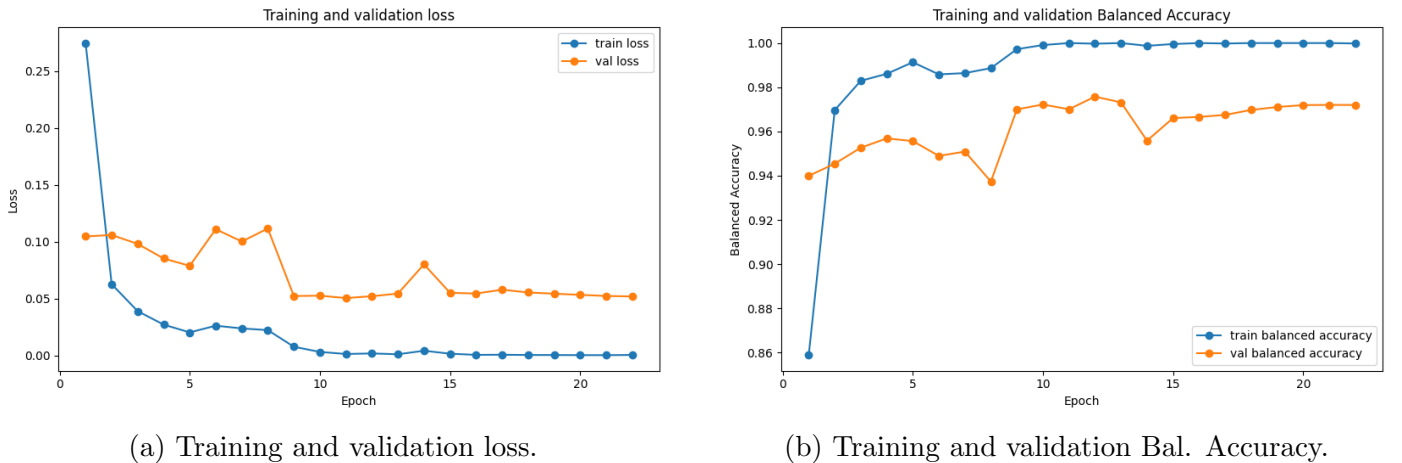


Figure 1: Training curves of the ResNet18 baseline.

5.1 Baseline error analysis

The baseline error analysis shows that the model is globally strong, but not uniformly reliable across classes. Plastic is the weakest class, with recall equal to 0.9191: 159 out of 173 validation samples are correctly classified, while most errors are confusions with Metal and Glass. This indicates a material-ambiguity problem, since plastic objects can be transparent, reflective or rigid, and can therefore resemble glass or metal objects.

Metal presents a complementary issue. Its recall is high, equal to 0.9805, but its precision is lower, equal to 0.9264, meaning that several non-metal samples are predicted as Metal. Therefore, the remaining errors are concentrated in a small set of visually ambiguous material classes rather than being uniformly distributed.

Since the best checkpoint was reached at epoch 12 and later epochs did not improve validation Balanced Accuracy, longer training was not considered the main solution. The following experiments were therefore targeted: augmentation was used to test robustness to visual variability, weighted CrossEntropy to address class imbalance, and letterbox padding to evaluate whether preserving aspect ratio could improve material discrimination.

True / Pred.	Bat.	Cloth.	Glass	Metal	Org.	Paper	Plastic	Undiff.
Battery	188	0	0	0	0	1	0	0
Clothing	0	1458	1	0	0	1	0	0
Glass	0	0	393	3	0	0	2	1
Metal	0	0	0	151	0	1	2	0
Organic	0	3	0	0	194	0	0	0
Papery	2	1	2	3	0	380	0	0
Plastic	0	1	5	6	0	2	159	0
Undifferentiated	1	2	0	0	1	0	1	134

Table 3: Raw confusion matrix for the ResNet18 baseline. Rows represent true classes, columns represent predicted classes.

6 Controlled ResNet18 Experiments

After the ResNet18 baseline, three controlled experiments were performed to test whether targeted changes could improve the remaining weaknesses of the model. The baseline already achieved high validation performance, so the goal was to evaluate one modification at a time and measure its effect on Balanced Accuracy, per-class recall and computational cost.

The baseline is denoted as B0. The following experiments are denoted as B1, B2 and B3. All experiments used the same train/validation split, label mapping, input size, batch size and ResNet18 architecture initialized with ImageNet-pretrained weights. The best checkpoint was always selected according to validation Balanced Accuracy. This setup makes the comparison controlled, because each difference in performance can be mainly attributed to the specific modification introduced in that experiment.

6.1 B1: Light Augmentation

The first controlled modification was applied only to the training data pipeline. The motivation came from the baseline error analysis: most residual errors involved visually ambiguous material classes, especially Plastic confused with Metal or Glass. Therefore, B1 tested whether exposing the model to moderate visual variations during training could improve robustness on such borderline cases.

The validation preprocessing was kept deterministic and identical to the baseline. This was necessary to keep validation metrics directly comparable and to measure performance on unchanged validation images. The training transform was extended with random resized cropping, horizontal flipping, small rotations and limited color jitter. These transformations were selected because they preserve the object label while introducing moderate changes in position, scale, orientation and illumination.

The result was informative but not beneficial overall. Plastic recall improved slightly, from 0.9191 to 0.9306, and Undifferentiated recall also increased. However, this improvement came at the cost of a clear degradation in Metal recall, which decreased from 0.9805 to 0.9416. As a consequence, the validation Balanced Accuracy dropped from 0.9758 to 0.9716.

This suggests that the chosen augmentation policy helped some visually variable samples but also made the model less stable on Metal. Since the main metric decreased, B1 was not selected as an improvement over the baseline. However, it was useful to show that simple augmentation was not

automatically beneficial for this dataset and that stronger robustness did not necessarily translate into better balanced performance.

6.2 B2: Weighted CrossEntropy

The second controlled modification addressed the class imbalance observed in the dataset. Since Clothing represents almost half of the images, a standard CrossEntropy loss can be dominated by the most frequent classes.

B2 kept the same architecture, preprocessing and training setup of the baseline, and changed only the loss function. Standard CrossEntropy was replaced with class-weighted CrossEntropy. Class weights were computed from the training split only, using:

$$w_c = \frac{N}{C \cdot n_c}$$

where w_c is the weight of class c , N is the number of training images, C is the number of classes, and n_c is the number of training samples of class c . This gives larger weights to under-represented classes and smaller weights to dominant classes, increasing the contribution of minority-class errors during training.

B2 produced the best ResNet18 validation Balanced Accuracy, although the improvement over the baseline was small. Balanced Accuracy increased from 0.9758 to 0.9764, and Plastic recall improved from 0.9191 to 0.9422. This confirms that weighting the loss helped the weakest class of the baseline. However, the improvement was not uniform: macro F1 slightly decreased and validation loss increased, which is expected because weighted and unweighted CrossEntropy optimize different objectives and their loss values are not directly comparable.

Overall, B2 was the most useful modification because it improved the main metric and directly addressed the imbalance problem. Nevertheless, the gain was marginal, suggesting that ResNet18 was already close to saturation on this split and that larger improvements would probably require a stronger architecture rather than further tuning of the same baseline.

6.3 B3: Letterbox Padding

The third experiment modified the preprocessing geometry. Instead of directly resizing each image to 224×224 , images were first padded to a square canvas and then resized. This was introduced because the EDA showed heterogeneous aspect ratios, and direct resizing could distort object shapes.

The padding color was set to the ImageNet mean RGB value, so that the padded area becomes approximately neutral after normalization:

```
PadToSquare(fill=(124, 116, 104))
Resize((224, 224))
ToTensor()
Normalize(ImageNet mean/std)
```

This preprocessing was motivated by the idea that shape and proportions may help distinguish visually similar classes such as Plastic, Glass, Metal and Papery. However, the result was nearly neutral: letterbox padding did not improve global Balanced Accuracy and introduced additional preprocessing cost. Therefore, direct resizing was not identified as a main limitation of the ResNet18 baseline.

6.4 Discussion

The controlled ResNet18 experiments show that the baseline was already a very competitive configuration. Light augmentation, weighted CrossEntropy and letterbox padding each affected the model in a different way, but none of them produced a substantial global improvement. B2 was the most coherent improvement with respect to the project metric, because it directly targeted class imbalance and achieved the best Balanced Accuracy among the ResNet18 variants. Nevertheless, its improvement over

the baseline was marginal. This motivated the following architectural experiments with EfficientNet, aimed at verifying whether a different CNN family could provide a better trade-off between Balanced Accuracy, per-class recall, memory usage and inference speed.

Experiment	Main change	Best epoch	Best val loss	Min val loss	Val Bal. Acc.	Δ Bal. Acc.
B0 ResNet18 baseline	Standard resize, CE loss	12	0.0521	0.0505 ep. 11	97.58%	–
B1 Light augmentation	Training augmentation	29	0.0592	0.0592 ep. 29	97.16%	-0.42 pp
B2 Weighted CE	Class-weighted loss	12	0.0718	0.0718 ep. 12	97.64%	+0.06 pp
B3 Letterbox padding	Pad-to-square before resize	12	0.0564	0.0543 ep. 22	97.56%	-0.02 pp

Table 4: Comparison between Resnet18 experiments: baseline vs others

Experiment	Battery	Clothing	Glass	Metal	Organic	Papery	Plastic	Undiff.
B0 ResNet18 baseline	99.47%	99.86%	98.50%	98.05%	98.48%	97.94%	91.91%	96.40%
B1 Light augmentation	98.41%	99.86%	98.50%	94.16%	97.46%	98.71%	93.06%	97.12%
B2 Weighted CE	98.94%	99.52%	97.74%	97.40%	98.98%	97.16%	94.22%	97.12%
B3 Letterbox padding	98.41%	99.79%	98.25%	98.70%	98.98%	97.42%	92.49%	96.40%

Table 5: Per-class recall comparison for the ResNet18 experiments.

7 EfficientNet-B0 Full Fine-Tuning

After the controlled ResNet18 experiments, EfficientNet-B0 was introduced to verify whether changing the CNN architecture could improve the accuracy-efficiency trade-off. The previous ResNet18 variants produced only marginal improvements, therefore the next experiment kept the setup controlled and replaced only the backbone architecture.

The experiment, denoted as `efficientnet_b0.b0.full_ft`, used the same train/validation split, label mapping, input size, batch size, loss function, optimizer and checkpoint criterion as the ResNet18 baseline. The model was initialized with ImageNet-pretrained weights, fully fine-tuned on the waste dataset, and adapted by replacing the final classifier with an 8-output layer. Images were converted to RGB, resized to 224×224 , converted to tensors and normalized using the ImageNet statistics expected by EfficientNet-B0. No data augmentation was applied, so the experiment isolates the effect of the architecture.

EfficientNet-B0 was selected because it is a compact CNN designed to balance accuracy and computational cost. Differently from ResNet18, which is based on residual blocks, EfficientNet uses compound scaling to coordinate network depth, width and resolution. This made it a suitable candidate after ResNet18: still compatible with the project constraints, but potentially more effective in feature extraction.

Table 6: Comparison between ResNet18 baseline and EfficientNet-B0 full fine-tuning.

Experiment	Val loss	Val Acc.	Val Bal. Acc.	Macro F1	Battery Rec.	Glass Rec.	Metal Rec.	Plastic Rec.	FPS
ResNet18 baseline	0.0521	98.64%	97.58%	97.70%	99.47%	98.50%	98.05%	91.91%	297.0
EfficientNet-B0	0.0399	98.71%	98.01%	97.87%	99.47%	97.74%	95.45%	96.53%	263.9
Difference	-0.0122	+0.06 pp	+0.43 pp	+0.17 pp	0.00 pp	-0.76 pp	-2.60 pp	+4.62 pp	-33.1

EfficientNet-B0 reached its best checkpoint at epoch 22, with a validation Balanced Accuracy of 98.01%. Compared with ResNet18, it improved the main metric by 0.43 percentage points. This suggests that the architectural change was more effective than the previous local modifications applied to ResNet18.

The improvement was not uniform across all classes. Plastic recall increased from 91.91% to 96.53%, while Metal recall decreased from 98.05% to 95.45%. Therefore, EfficientNet-B0 reduced the main weakness of the baseline, but introduced a small trade-off on Metal.

The confusion matrix shows that errors are mostly concentrated in visually ambiguous pairs. The most common error is mistaking glass for plastic, with 6 cases, followed by mistaking metal for battery, with 4 cases. Other minor confusions involve plastic and glass, glass and metal, and plastic and paper. These errors are likely due to the ambiguous photos in the dataset, where pairs may share similar visual cues across different material classes.

From a computational point of view, the model processed the 3,099 validation images in 11.74 seconds, reaching 263.87 FPS with 481.90 MB of peak GPU memory on CUDA. Thus, EfficientNet-B0 satisfies the memory and inference constraints, although it is slower than the ResNet18 baseline.

Overall, EfficientNet-B0 provided a better accuracy-oriented result than ResNet18, mainly due to the improvement on Plastic and the higher Balanced Accuracy. This motivated the next experiment with Weighted CrossEntropy on EfficientNet-B0, to test whether imbalance-aware training could further improve per-class recall on top of the stronger architecture.

8 EfficientNet-B0 with Weighted CrossEntropy

After the EfficientNet-B0 baseline, a second EfficientNet experiment was performed using class-weighted CrossEntropy. The architecture, preprocessing, optimizer, input size and batch size were kept unchanged. Therefore, the main difference was the loss function, introduced to give more importance to under-represented classes during training.

Class weights were computed only from the training split, using the same strategy already tested in the ResNet18 experiments. This keeps the validation set independent and makes the experiment coherent with the main metric, Balanced Accuracy, which depends on the recall of each class.

Table 7: Comparison between ResNet18 and EfficientNet-B0 using Weighted CrossEntropy.

Experiment	Best epoch	Val Bal. Acc.	Glass Rec.	Metal Rec.	Plastic Rec.	Undiff. Rec.	FPS
ResNet18 Weighted CE	12	97.64%	97.74%	97.40%	94.22%	97.12%	283.7
EfficientNet-B0 Weighted CE	16	98.22%	98.25%	97.40%	95.38%	99.28%	281.3
Difference	+4	+0.58 pp	+0.51 pp	0.00 pp	+1.16 pp	+2.16 pp	-2.4

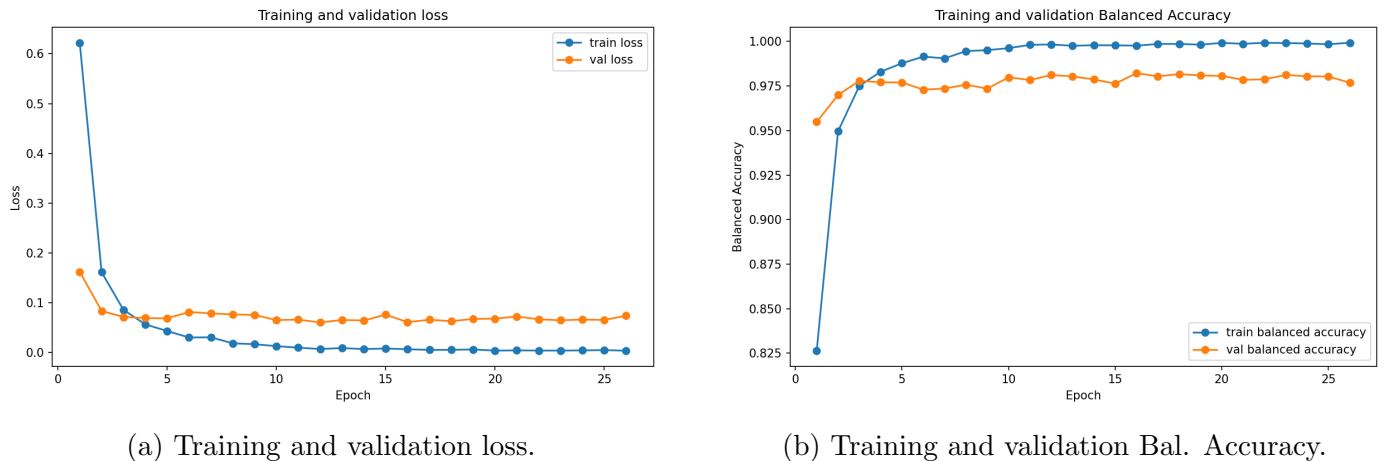


Figure 2: Training curves of EfficientNet-B0 with Weighted CrossEntropy.

On the 3,099 validation images, EfficientNet-B0 reduces the total number of misclassifications from 49 (ResNet18) to 37.

This global improvement is largely driven by a substantial error reduction in specific classes: misclassifications for Papery dropped from 11 to 6, and errors for Undifferentiated dropped from 4 to 1. Minor improvements were also observed for Glass and Plastic. This comparison isolates the architectural effect under the same imbalance-aware loss. Both models use Weighted CrossEntropy, but EfficientNet-B0 achieves a higher validation Balanced Accuracy than ResNet18. The improvement

is mainly due to better recall on Glass, Plastic and Undifferentiated, while Metal recall remains unchanged. The inference speed is almost the same, with EfficientNet-B0 being only slightly slower. Overall, EfficientNet-B0 with Weighted CrossEntropy provides the best trade-off between balanced validation performance and computational cost.

The training curves show fast convergence: validation performance improves early and then stabilizes, while training loss continues to decrease. For this reason, checkpoint selection based on validation Balanced Accuracy remains important. The best model was obtained at epoch 16.

9 Cross-validation protocol

After the fixed-split experiments, we applied 5-fold cross-validation only to the three most relevant configurations: ResNet18 with Weighted CrossEntropy, EfficientNet-B0 with standard CrossEntropy, and EfficientNet-B0 with Weighted CrossEntropy. We did not repeat cross-validation for all preliminary experiments, because this would have increased the computational cost without adding useful information for model selection.

The goal of this phase was to obtain a more stable estimate of validation performance across different data partitions. In particular, cross-validation allowed us to verify whether the lower recall observed for minority or visually ambiguous classes, such as Metal, Plastic and Glass, was caused by an unfavorable hold-out split or represented a systematic difficulty of the task.

For each configuration, the final comparison was based on the mean and standard deviation of Balanced Accuracy across the five folds. A representative fold was then selected only for detailed analysis, confusion matrix visualization and computational benchmarking.

10 ResNet18 Weighted CrossEntropy 5-fold cross-validation

This experiment evaluated the best ResNet18 fixed-split configuration under the 5-fold cross-validation protocol. The same ResNet18 architecture, input size, batch size, optimizer and Weighted CrossEntropy loss were kept. For each fold, the model was initialized from ImageNet-pretrained weights and the best checkpoint was selected according to validation Balanced Accuracy.

Table 8: Normalized confusion matrix of the ResNet18 Weighted CE representative fold (fold 4).

True \	Bat	Clo	Gla	Met	Org	Pap	Pla	Und
Bat	0.98	0.00	0.01	0.00	0.00	0.02	0.00	0.00
Clo	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
Gla	0.00	0.00	0.97	0.01	0.00	0.00	0.01	0.00
Met	0.04	0.01	0.02	0.91	0.00	0.01	0.01	0.00
Org	0.00	0.02	0.00	0.00	0.98	0.00	0.00	0.00
Pap	0.01	0.00	0.00	0.01	0.00	0.98	0.00	0.00
Pla	0.00	0.02	0.03	0.01	0.00	0.03	0.92	0.00
Und	0.00	0.00	0.01	0.00	0.01	0.01	0.00	0.98

Bat = Battery, Clo = Clothing, Gla = Glass, Met = Metal, Org = Organic, Pap = Papery, Pla = Plastic, Und = Undifferentiated.

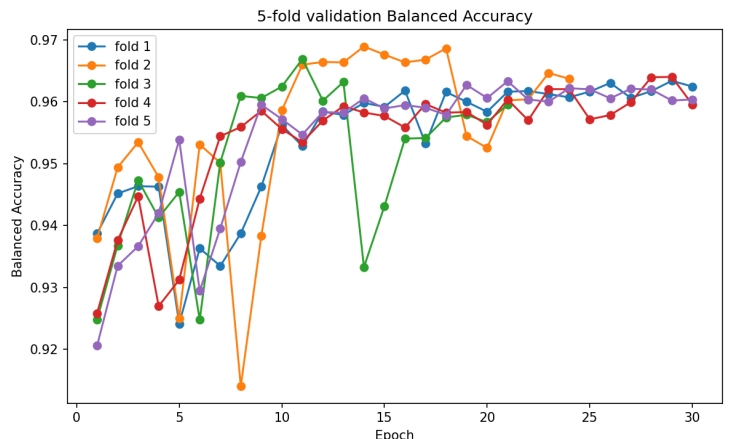


Figure 3: Validation Balanced Accuracy across the five folds.

The mean cross-validation Balanced Accuracy was 96.53%, with a standard deviation of 0.25 percentage points. This indicates that the ResNet18 Weighted CrossEntropy configuration is stable across different validation partitions. The representative fold was fold 4, selected because its Balanced Accuracy was the closest to the cross-validation mean. This fold achieved 96.40% Balanced Accuracy and 97.84% accuracy at epoch 29.

The selected fold confirms the same behavior observed in the previous ResNet18 experiments. The model performs well on Clothing, Organic, Papery, Battery and Undifferentiated, while the weakest classes remain Metal and Plastic. In particular, Metal recall is 90.91% and Plastic recall is 91.91%, showing that most residual errors are still concentrated in visually ambiguous material classes.

From a computational point of view, the selected ResNet18 fold remains very lightweight. It processed 3,103 validation images in 9.77 seconds, reaching 317.58 FPS with 404.37 MB of peak GPU memory on CUDA. Therefore, the model largely satisfies the project constraints on inference memory and speed.

However, the per-class results confirm that Metal and Plastic remain the weakest classes even when the validation partition changes. This suggests that their lower recall is not only due to a single unfavorable split, but also to intrinsic visual ambiguity with other material classes. For this reason, we evaluated EfficientNet-B0 under the same cross-validation protocol, to verify whether a stronger architecture could improve these residual errors while preserving the computational constraints.

11 EfficientNet-B0 baseline with 5-fold cross-validation

This experiment evaluated EfficientNet-B0 with standard CrossEntropy under the same 5-fold protocol used for ResNet18. The purpose was to verify whether the improvement observed in the fixed-split experiments was stable across different validation partitions.

The architecture was fully fine-tuned from ImageNet-pretrained weights, using the same input size, optimizer, scheduler, early stopping strategy and model-selection metric adopted in the previous experiments. Therefore, the main difference with respect to the ResNet18 cross-validation experiment is the CNN backbone.

Table 9: Normalized confusion matrix of the selected representative fold (fold 3).

True \	Bat	Clo	Gla	Met	Org	Pap	Pla	Und
Bat	0.97	0.00	0.00	0.02	0.00	0.00	0.01	0.00
Clo	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
Gla	0.00	0.00	0.96	0.00	0.00	0.00	0.03	0.00
Met	0.01	0.00	0.01	0.96	0.00	0.01	0.01	0.00
Org	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00
Pap	0.00	0.00	0.00	0.01	0.00	0.98	0.01	0.00
Pla	0.00	0.00	0.01	0.01	0.00	0.01	0.98	0.00
Und	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.99

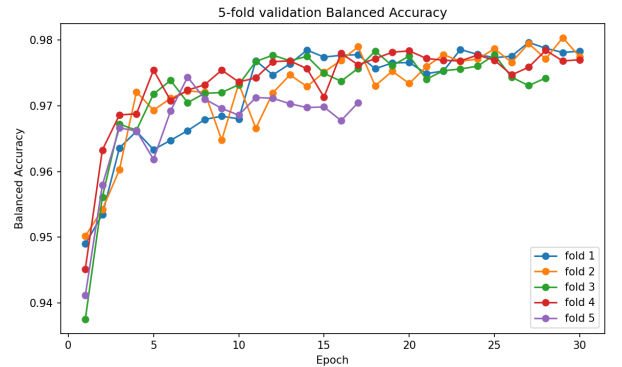


Figure 4: Validation Balanced Accuracy across the five folds.

The 5-fold cross-validation produced stable results. The mean validation Balanced Accuracy was 97.82%, with a standard deviation of 0.23 percentage points. The best Balanced Accuracy values across the five folds were 97.97%, 98.04%, 97.83%, 97.85% and 97.44%, showing limited variation among validation partitions.

The representative model was the checkpoint from fold 3, because its Balanced Accuracy was closest to the cross-validation mean. This fold reached 97.83% Balanced Accuracy and 98.29% standard accuracy, with the best checkpoint obtained at epoch 18.

The per-class analysis confirms that EfficientNet-B0 performs well across all classes, including minority classes. Clothing remains the easiest class, while the main residual difficulties are concentrated on Plastic, Metal and Glass. This confirms that these classes are systematically more difficult, rather than being weak only in the original hold-out split.

From a computational point of view, EfficientNet-B0 also satisfies the project constraints. On the selected validation fold, inference on 3,103 images required 9.91 seconds, corresponding to about 313 FPS, with a peak GPU memory usage of only 257 MB. Therefore, EfficientNet-B0 provides a better accuracy-memory trade-off than ResNet18 and remains suitable for deployment.

12 EfficientNet-B0 Weighted CE 5-fold CV

The last experiment combines the best-performing architecture with an imbalance-aware training objective. Starting from the EfficientNet-B0 5-fold CrossEntropy baseline, we kept the same architecture, input size, optimizer, scheduler, training protocol and cross-validation procedure. The only relevant modification was the replacement of standard CrossEntropy with Weighted CrossEntropy.

The class weights were computed independently inside each fold, using only the corresponding training portion. This follows the same weighting strategy already described in the previous Weighted CrossEntropy experiments and avoids validation leakage.

The goal was to verify whether the imbalance-aware loss could improve mean Balanced Accuracy and minority-class recall without changing inference cost, since the loss function affects only training and not the final network architecture.

Table 10: Normalized confusion matrix of the selected representative fold (fold 5).

True \	Bat	Clo	Gla	Met	Org	Pap	Pla	Und
Bat	0.98	0.00	0.00	0.01	0.00	0.01	0.00	0.00
Clo	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
Gla	0.00	0.00	0.98	0.00	0.00	0.00	0.01	0.00
Met	0.01	0.00	0.01	0.95	0.01	0.00	0.02	0.01
Org	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00
Pap	0.00	0.00	0.00	0.01	0.00	0.98	0.00	0.00
Pla	0.00	0.00	0.01	0.00	0.00	0.00	0.97	0.02
Und	0.01	0.00	0.00	0.00	0.01	0.00	0.01	0.97

Bat = Battery, Clo = Clothing, Gla = Glass, Met = Metal, Org = Organic, Pap = Papery, Pla = Plastic, Und = Undifferentiated.

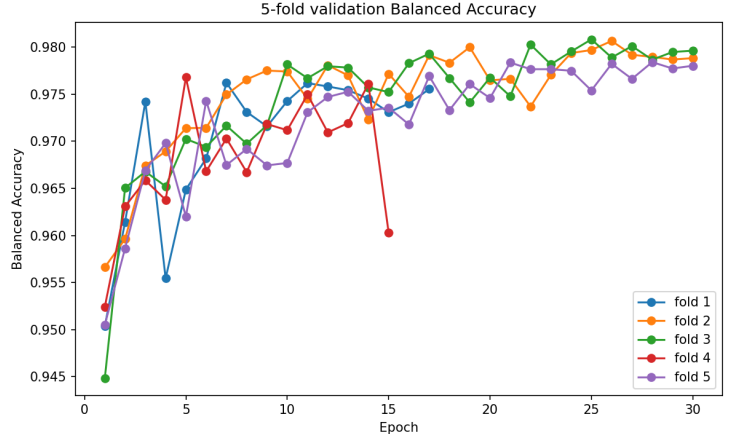


Figure 5: Validation Balanced Accuracy across the five folds.

The EfficientNet-B0 Weighted CE experiment achieved a mean 5-fold Balanced Accuracy of $97.86\% \pm 0.21\%$, showing stable performance across folds. The representative model was fold 5, selected because its Balanced Accuracy was close to the cross-validation mean. This fold reached 98.52% validation accuracy, 97.84% Balanced Accuracy and 97.43% macro F1.

Compared with EfficientNet-B0 standard CrossEntropy, the improvement in mean Balanced Accuracy is very small, from 97.82% to 97.86%. However, Weighted CE preserves the same inference cost because the architecture is unchanged. The selected model reached 319.04 FPS with only 257.15 MB of peak GPU memory.

Therefore, this experiment shows that Weighted CE can be adopted without any deployment penalty, although the gain over standard CrossEntropy remains marginal.

13 Final model selection

The final selected model is EfficientNet-B0 with Weighted CrossEntropy evaluated with 5-fold cross-validation. This configuration achieved the best mean Balanced Accuracy, $97.86\% \pm 0.21\%$, while keeping the same inference cost as the standard EfficientNet-B0 CrossEntropy model.

The difference with standard CrossEntropy is marginal, since the mean Balanced Accuracy increases only from 97.82% to 97.86%. For this reason, the choice is not based on a large performance gap. Weighted CE is selected because it obtains the best mean Balanced Accuracy, is slightly more stable across folds, and is more coherent with the class imbalance of the dataset and with Balanced Accuracy as the main project metric.

Since Weighted CE affects only the training loss, the final model preserves the same architecture, memory usage and inference speed of EfficientNet-B0. The selected representative fold reached 98.52% validation accuracy, 97.84% Balanced Accuracy and 97.43% macro F1, with 319.04 FPS and 257.15 MB of peak GPU memory. Therefore, EfficientNet-B0 Weighted CE was selected as the final model, while EfficientNet-B0 standard CrossEntropy remains a valid simpler alternative.